A Field Guide to Integration and System Tests

There are a lot of different names for a lot of specific kinds of tests:

- The Rails core testing library defines two similar types of end-to-end tests: *integration tests* and *system tests*. System tests, which were new in Rails 5.1, require the Capybara gem to simulate user interaction with the application and to query application output.[1] Integration tests, which have always been a part of Rails, also simulate user interaction, but use a Rails-specific API that is less complete than Capybara. (To make it more confusing, you can add Capybara for use in integration tests if you want.)
- RSpec defines basically the same kind of tests but gives them different names. RSpec has *system* tests that are wrappers around the Rails core system tests. Historically, most RSpec users wrote *feature* specs, which are end-to-end tests that typically use Capybara. RSpec also has *request* specs, which are basically wrappers around Rails integration tests.

In addition, both RSpec and Rails/Minitest provide tools for specifically testing controllers, views, mailers, background jobs, and helpers. The following table helps sort out the naming.

Test Input	Test Evaluates	Common Name	Minitest Name	RSpec Name
User clicks via Capybara	HTML output	Integration	n/a	System (formerly feature)
User clicks via Capybara with JavaScript	HTML output	Integration	System	System (formerly feature)
URL for routing	HTML output	Integration	Integration	Request
Controller and action	Controller behavior	Controller	Controller	Controller

In RSpec, whether the test uses a JavaScript driver depends not on the kind of test you write, but on the driver you configure when you start the test. My best understanding of expected practice currently is that the Rails core team recommends using Capybara and system tests only if you're also evaluating code that requires JavaScript, and otherwise recommends the non-Capybara integration tests. The RSpec-Rails team, on the other hand, recommends the use of Capybara in system tests whether the test uses JavaScript or not.

1. https://github.com/teamcapybara/capybara